Observable Compliance, Hidden Distortions: Cropland Protection under Incomplete Monitoring

JMP

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Latest slides

Latest paper

When Policy Enforcement Is Decentralized

- The **center** sets **multiple targets** for **local governments**.
- **Multi-task local bureaucrats** are **promotion-driven**: target performance shapes their **careers**.
- With **asymmetric information**, local governments may deviate.

$\Rightarrow$ The center **strengthens monitoring** of previously neglected tasks.

$\Rightarrow$ When **monitoring is incomplete**, tighter supervision may **distort** behavior rather than correct it.

RQ

- When the center tightens supervision on what it can observe, does it induce **within-task distortion** on what it cannot observe?

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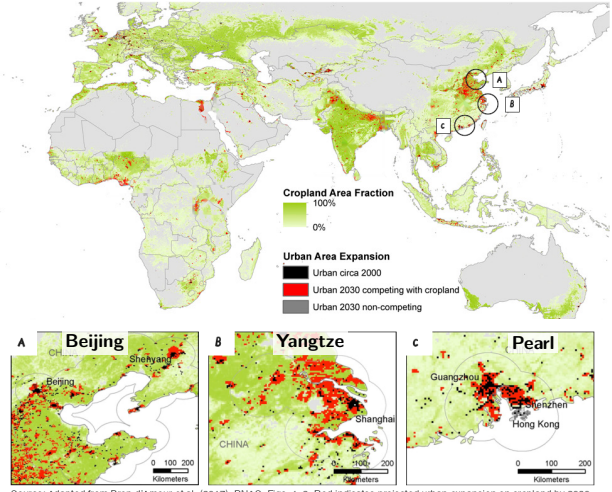
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Cropland loss from urban expansion



Source: Adapted from Bren d'Amour et al. (2017), PNAS, Figs. 1–2. Red indicates projected urban expansion on cropland by 2030.
Panels: A = Beijing-Tianjin-Hebei; B = Yangtze River Delta; C = Pearl River Delta.

China's Policy Response: Cropland Occupation–Compensation Balance

Central government's objectives

- **Economic growth (GDP)**: urban expansion requires construction land.
- **Food security**: protect cropland to sustain agricultural production.

Cropland Occupation–Compensation Balance Policy

When cropland is converted to construction land, local governments must compensate by creating an **equal amount** of cropland elsewhere.

Incentive tension

- **Multitask bureaucrats:** GDP (urban expansion) $\leftrightarrow$ cropland protection.
- **Within-task distortion** on cropland protection: quantity (monitoring strengthened after 2017) vs. quality (monitoring not strengthened).

Research Overview

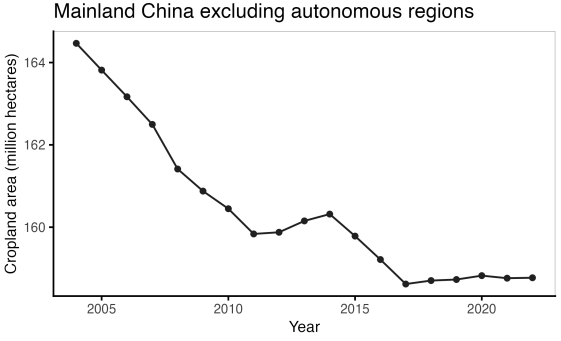
- **RQ**: When cropland monitoring becomes more stringent on *area* but not *quality*, does local implementation distort land allocation?
- **Background**
  - 2017 policy tightening: stricter enforcement of cropland area targets.
  - Jiangsu: high conflict between agricultural responsibility and economic-development pressure.
- **Data**
  - Satellite imagery + Admin data.
  - Unit of analysis: 500×500 m grid.
- **Method**
  - **DiD**: high- versus low-pressure counties before/after 2017.
  - **Spatial RD-DID**: nearby grid cells across administrative borders.
  - Pressure = predicted compensation burden on a county's rural land reservoir.
- **Results**
  - Quantity ↑, but new cropland is **less suitable** and **used less** (fallow ↑).
  - Quality losses go beyond exhausting good land: **distorted reclamation choices** toward **less-monitored** sites; larger losses under stronger **career incentives**.

Related Literature

- **Principal-Agent Problem.** Holmstrom and Milgrom (1991); Baker (1992); Dixit (2002); Courty and Marschke (2004); Bandiera et al. (2021); Chen et al. (2018); He et al. (2020); Cao et al. (2026); Hong and Huang (2025); Fan et al. (2025); Fang et al. (2023); Axbard and Deng (2024); Fisman et al. (2025).
- **Decentralization and Local Incentives.** Montinola et al. (1995); Jin et al. (2005); Zhang (2006); Xu (2011); Li and Zhou (2005); Zhang et al. (2018); Wang et al. (2020); Wang and Yang (2025); Luo et al. (2025).
- **Land Regulation in China.** Cai et al. (2017); Henderson et al. (2022); Fu et al. (2021); Kong et al. (2024); Fang et al. (2026); Lu et al. (2026); Hong et al. (2025); Chen and Li (2025); Xing et al. (2026); Yu (2026).
- **Land Misallocation and Agricultural Productivity.** Restuccia and Santaeulalia-Llopis (2017); Adamopoulos and Restuccia (2014); Adamopoulos et al. (2022); Chari et al. (2021); Chen et al. (2022); Chen et al. (2023); Adamopoulos et al. (2024).
- **Contribution: First evidence on within-task distortion under incomplete monitoring**

Background: Cropland Occupation–Compensation Balance

- Long-standing policy[a] – strengthened in 2017
- Core rule: occupy one unit, compensate one unit[b]
- Implementation:
  Upper-level government → lower-level government
  assigns **planned land use quota (cap)**
- **Planned land use quota**:
  new construction land converted from agricultural land
  = **cropland-compensation obligation**



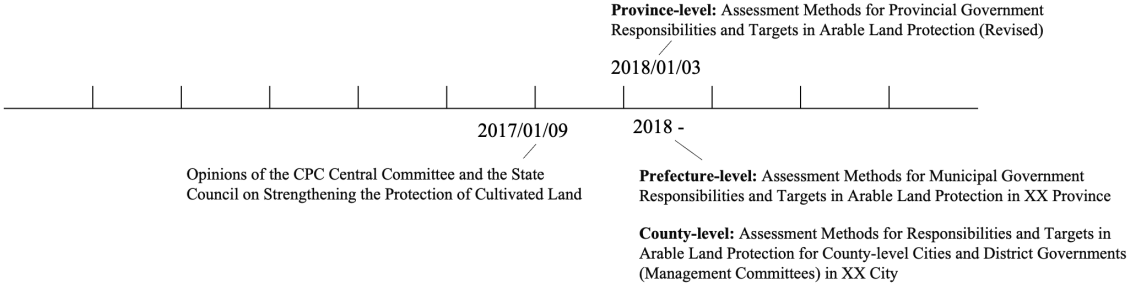
Full mainland sample

[a] (in Chinese: 耕地占补平衡政策)

[b] Compensation cropland → **permanent basic farmland**: quality shortfalls locked in.

2017–2018 Policy Tightening

- **Cropland Occupation–Compensation Balance**: 2017–2018 revisions assessed local cropland protection more stringently.[1]



[1] Document of 2018 policy

Background: Regulation and Local Implementation

Implementation of the balance policy

- **Central government's objective**: quantity-based and productive-capacity-centered.
- **Implementation distortion**: ✓ quantity, × quality.

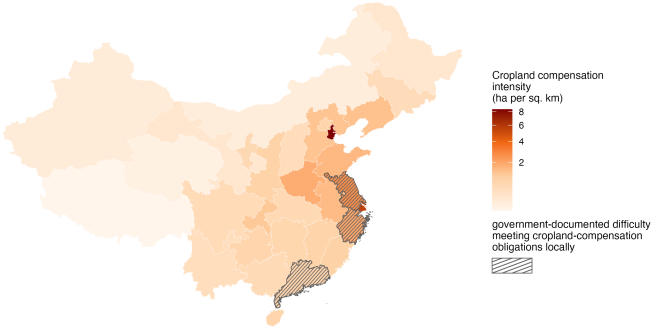


Local implementation: forest → terrace.

- **Multitask bureaucracy**: economic growth + cropland protection.
- **Land-use regulation**: government designates land for construction or agricultural use.

Why Jiangsu?

- **Cropland-Compensation Intensity**$_p$ = obligation$_p$ / provincial land area$_p$
- **Not all provinces face a binding land-use conflict.**
- **Jiangsu is a sharp case:** a major agricultural and economic province, mostly flat.



Measurement of Key Outcomes

Data: satellite imagery + administrative data.

Observation unit: 500 m $\times$ 500 m grid cell.

1. **Cropland share:** land-cover data; fraction of 30 m pixels classified as cropland.[1]
2. **Quality / agricultural suitability:** soil-based suitability for cultivation (FAO 1976; FAO 2007).[2]
3. **Cultivation intensity:** annual Fractional Vegetation Cover (FVC).
   - *Fallow*: average annual FVC below 0.20 (bottom 25%).

Full data table

[1] Reclaimed cropland: a grid whose cropland share is low in $t-1$ and reaches a high enough level in both t and $t+1$.

[2] The source-level indicator $1(\text{score} \geq 3)$ is 0 for saline–alkali or other non-arable land; area-weighted aggregation gives each grid's suitable-land share.

Measurement of Cropland Protection Pressure

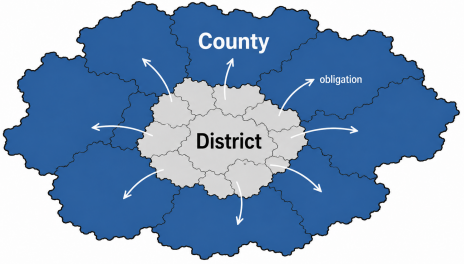
Cropland Occupation–Compensation Balance

(1) **Equal-area compensation (one-for-one).**[1]

(2) **Compensation is not necessarily local.**

- within the same county,
- if infeasible (e.g., urban core district), within the same prefecture (across counties)[2].

This creates a *capacity constraint* determined by the non-core (non-urban) area within a prefecture.



[1] 占一补一: a county converting cropland to construction land compensates with an equal amount of cropland elsewhere.

[2] Jurisdictions that cannot meet the balance requirement can purchase supplementary cropland indicators through a paid, auction-based transaction platform.

Measuring County-Level Cropland Protection Pressure

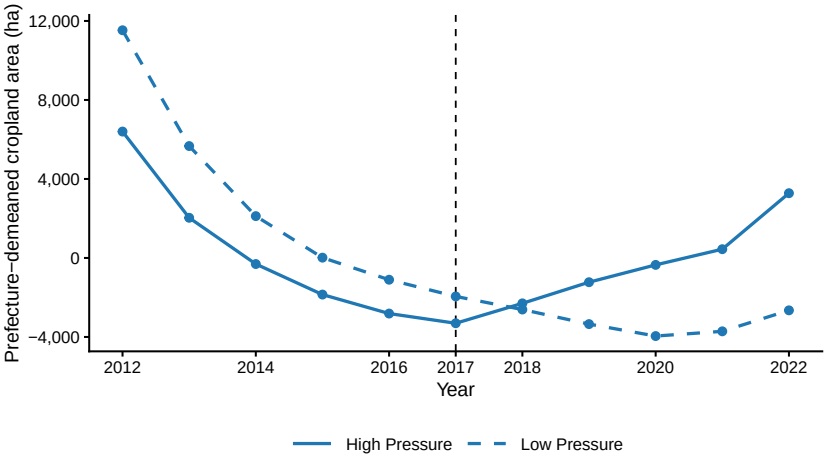
$$\text{Pressure}_c = \frac{\text{Obligation}_p}{\text{TotalLand}_p} \times \text{LandShare}_c \times 1000$$

- Obligation_p: planned compensation obligation of prefecture p's urban core.
- TotalLand_p: available land outside the urban core in prefecture p.[1]
- $\text{LandShare}_c = \frac{\text{AvailableLand}_c}{\text{TotalLand}_p}$, where AvailableLand_c is county c's available land.

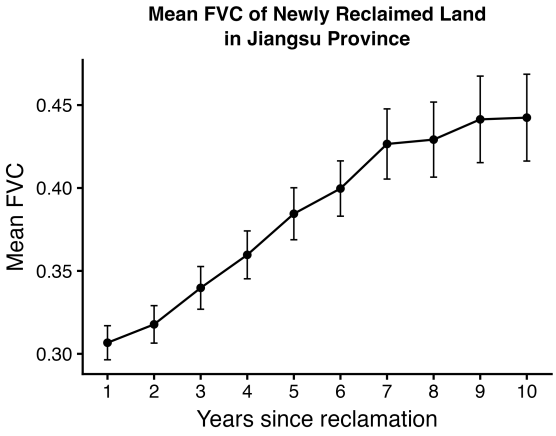
Jiangsu pressure map Pressure distribution

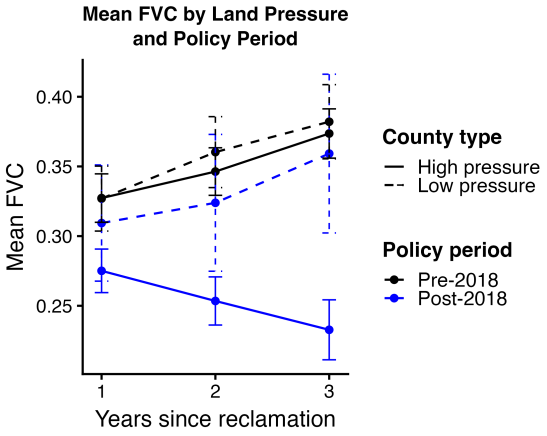
[1] Available land excludes pre-policy impervious and water shares.

Motivating Fact: Rural Cropland Area



Motivating Fact: Cultivation intensity and FVC





Identification Specification

(1) Event-study DID: cropland share of all grids

$$\text{CropShare}_{ict} = \sum_{k \neq -1} \beta_k \big(\text{Pressure}_c \times \text{Event}_{k,t}\big) + \mu_i + \lambda_t + \varepsilon_{ict}$$

- $\text{Event}_{k,t}$: event-time indicators relative to 2018 ($k = -1$: 2017 omitted).
- **FE:** grid (μ_i) + year (λ_t).

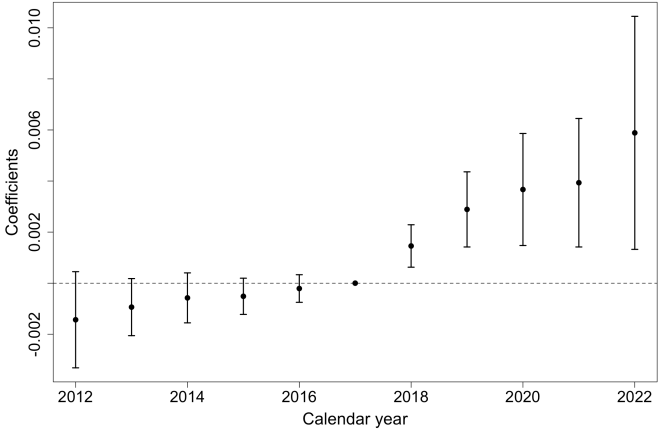
(2) Cohort DID: fallow & soil suitability of newly reclaimed land

$$Y_{i,s} = \beta\big(\mathbf{1}\{t_0(i) \geq 2018\} \times \text{Pressure}_c\big) + \alpha_c + \mu_{t_0(i)} + \delta_s + \varepsilon_{i,s}$$

- The outcomes are attributes of the newly reclaimed land — fallow and soil suitability undefined before reclamation $\Rightarrow$ cohort DID.
- **Cohort** $t_0(i)$ = grid i's reclamation year.
- **FE:** county (α_c) + cohort ($\mu_{t_0(i)}$) + years-since-reclamation (δ_s).

Conceptual framework Robustness checks

Cropland Share Expansion



- 1 s.d. Pressure$_c$ $\uparrow \Rightarrow$ **+0.16 p.p.** cropland share in 2018 ($\approx$ **13,700 ha**); $\approx$ **84%** of the pre-period annual cropland expansion.

  Relative to 2017; pre-period annual expansion = avg. gross cropland gain 2013–2017 in the non-core grid sample ($\approx$ 16,200 ha/yr).

Soil Suitability and Fallow Rate

	Soil suitability				$\mathbf{1}$(Fallow)	
	1995 soil data		2023 soil data			
	(1)	(2)	(3)	(4)	(5)	(6)
Pressure × Post	-0.051**	-0.057***	-0.032*	-0.037*	0.055***	0.056***
	(0.020)	(0.021)	(0.018)	(0.019)	(0.015)	(0.017)
Control Mean	0.593	0.593	0.573	0.573	0.198	0.200
County controls × year		✓		✓		✓
County & Cohort FE	✓	✓	✓	✓	✓	✓
Years-since-reclamation FE					✓	✓
Cluster level	County	County	County	County	County	County
Number of counties	52	52	52	52	54	53
Num. Obs.	939	939	939	939	3,244	3,222
R^2	0.452	0.457	0.490	0.491	0.232	0.230

A 1 s.d. increase in pressure after 2017:

- **Suitable-land share**: 9.6% lower (1995 soil data); 6.1% lower (2023 soil data).
- **Fallow rate**: 30.8% higher than baseline.

Columns (1), (3), (5); control means: 0.593, 0.573, 0.198.

Crop output per hectare Fallow event study

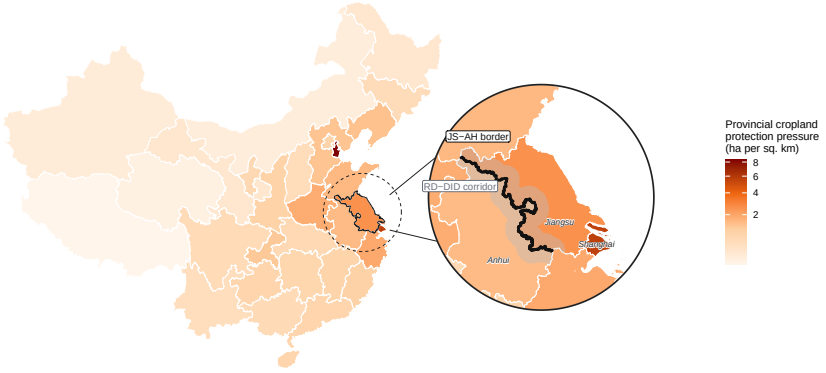
Robustness Checks

- **Parallel trends for the cultivation-intensity measure** — Event study of FVC
- **Alternative definitions of land pressure** — Appendix table
- **Sensitivity to reclaimed-land classification** — Appendix figures
- **Sensitivity to fallow definition** — Appendix figure

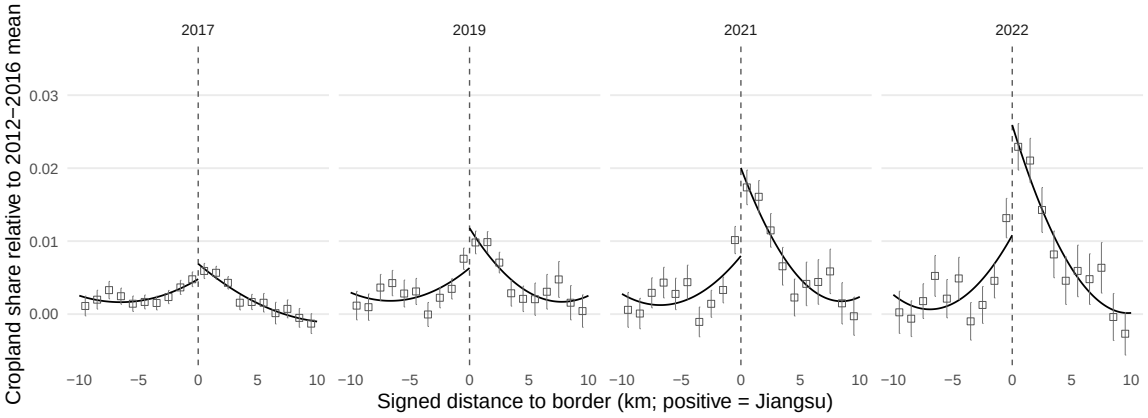
Back to specification

Spatial RD-DID

Compare pre-post change in outcome across the border.

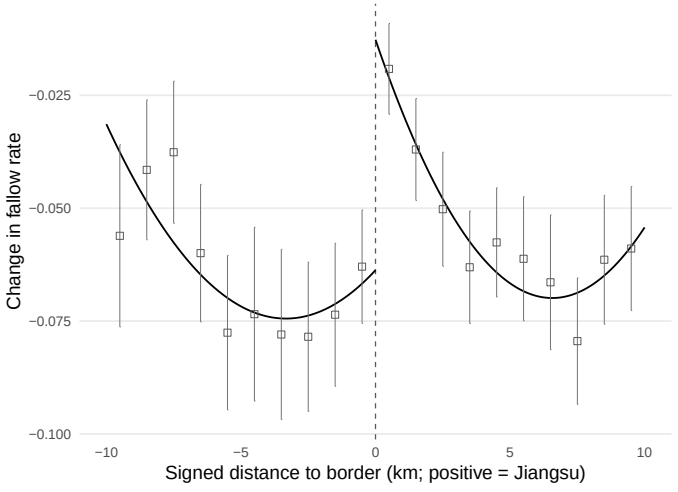


Spatial RD-DID: Δ Cropland Share (vs. 2012–2016 Mean) at the Jiangsu–Anhui Border



Placebo: assign policy to 2013/14 ⇒ no pre-period border gap

Spatial RD-DID: Δ Fallow Rate at the Jiangsu–Anhui Border



Spatial RD-DID: Δ Cropland Share and Δ Fallow Rate

	Main comparison			Robustness comparison		
	Jiangsu–Anhui			Jiangsu–Shanghai		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Δ Cropland share						
RD in Δ cropland share	0.006**	0.006**	0.007**	−0.027*	−0.027*	−0.029**
(Jiangsu − comparison side)	(0.003)	(0.003)	(0.003)	(0.014)	(0.015)	(0.012)
Panel B: Δ Fallow rate						
RD in Δ fallow rate	0.037***	0.038***	0.037***	−0.256***	−0.266***	−0.266***
(Jiangsu − comparison side)	(0.010)	(0.011)	(0.014)	(0.035)	(0.031)	(0.042)
County-pair FE	✓	✓	✓	✓	✓	✓
Kernel	Triangle	Epanech.	Uniform	Triangle	Epanech.	Uniform

Notes. Outcomes are 2018–2022 means minus 2012–2017 means. Local-linear RD-DID in pre-to-post changes within the ± 50 km corridor, with additive county-pair indicators and MSE-optimal bandwidths. Entries are robust bias-corrected estimates; robust standard errors clustered by county are in parentheses. Panel B uses complete-FVC ex ante marginal cells.

Additional Results and Alternative Explanations

Additional results

- New cropland is more isolated and fragmented.
- Greater conversion of water bodies and wetlands to cropland.

Full results table

Alternative explanations for higher fallow

- **Normal seasonality**: higher fallow persists in April–May and July–September. Evidence
- **Temporary land preparation**: no gap in years 1–2; a gap emerges in year 3. Evidence
- **Crop switching**: no clear shift among detected cultivation. Evidence

Mechanism: Non-Distortive, Costly Compliance or Distorted Choices?

Non-distortive, costly compliance

Quality falls because the best land has already been used.

Distorted choices

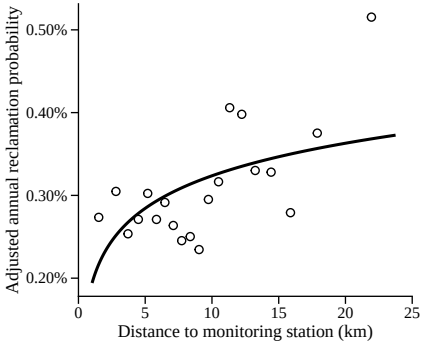
Local governments choose less suitable land even when better options remain.

Evidence for distorted reclamation choices

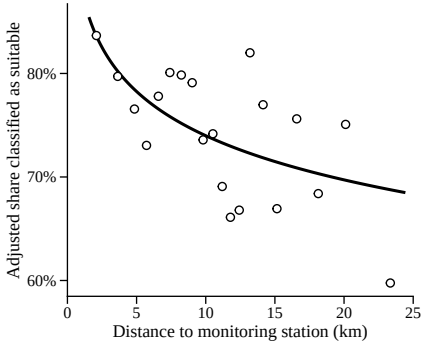
1. **Less monitoring**: post-policy reclamation moves toward weakly monitored, far-from-station land.
   ▸ Table ▸ Figure ▸ Robustness
2. **Lower costs at the expense of suitability**: higher pressure shifts reclamation toward low-cost, unsuitable land and away from high-cost, suitable land.
   ▸ Evidence
3. **Decomposing quality losses: compliance vs. distortion**. Predict quality from historical choices, assuming unchanged selection patterns.
   ▸ Decomposition

$$\underbrace{\text{Predicted quality} - \text{Actual quality}}_{\text{Distortion proxy}} = 73\text{–}78\% \times \underbrace{\text{Pre-policy quality} - \text{Actual quality}}_{\text{Total quality decline}}$$

Mechanism: Monitoring-Station Distance and Reclamation Choices



(a) Reclamation probability, 2018–22



(b) Suitable share of reclaimed land

- **Heterogeneity analysis** (Post $\times$ log Distance): the shift toward parcels **farther from monitoring stations strengthens after 2017**.

▸ Table ▸ Coef. stability ◂ Back to mechanism

Do Officials' Incentives Explain These Distortions? — Yes

	1995 soil data Suitable share (1)	2023 soil data Suitable share (2)
Pressure × Post	-0.097^{***}	-0.067^{**}
	(0.021)	(0.022)
	[0.006]	[0.089]
Incentive × Pressure × Post	$\mathbf{-0.058}^{***}$	$\mathbf{-0.067}^{***}$
	(0.013)	(0.021)
	[0.022]	[0.067]
Control mean	0.593	0.573
Party-secretary controls	✓	✓
County linear trends	✓	✓
County & cohort FE	✓	✓
Number of counties	52	52
Observations	939	939
R^2	0.498	0.532
Prefecture clusters	13	13

- **Hypothesis**: stronger promotion incentives favor visible acreage targets over land quality.
- **Promotion incentive** $= 1$ if the prefecture party secretary is ≤ 57 and the year precedes the 2020 national land inspection.

Notes. Suitable share: area-weighted source-pixel/component indicators for scores ≥ 3. Post: after 2017. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

▸ Economic growth pressure ▸ Agricultural losses and land revenue

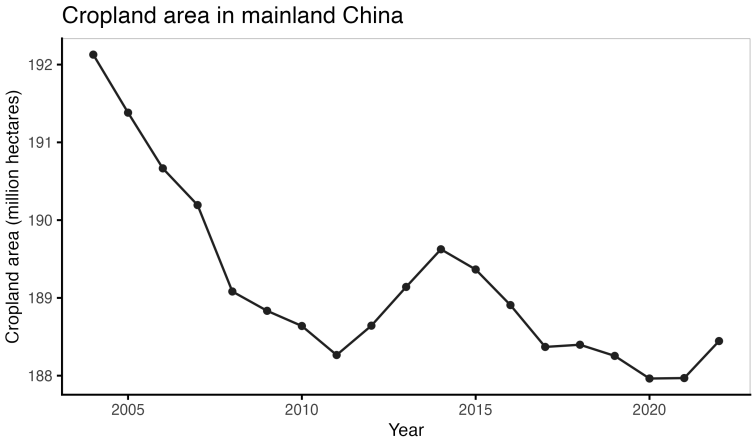
Wrap-Up

- Multitasking bureaucrats generate within-task distortions under incomplete monitoring.
- The policy targets area and productivity; enforcement strengthens mainly on area.
- Satellite-based evidence from DiD and spatial RD-DID designs.
- More cropland, but lower suitability and less intensive use.
- More fragmentation and conversion of water bodies and wetlands.
- **Distorted reclamation choices**: shifts toward less-monitored sites; larger quality losses under stronger career incentives.

Thank You!

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Full Mainland Cropland Area



Data source: CLCD county data. Mainland excludes Hong Kong, Macao, and Taiwan.

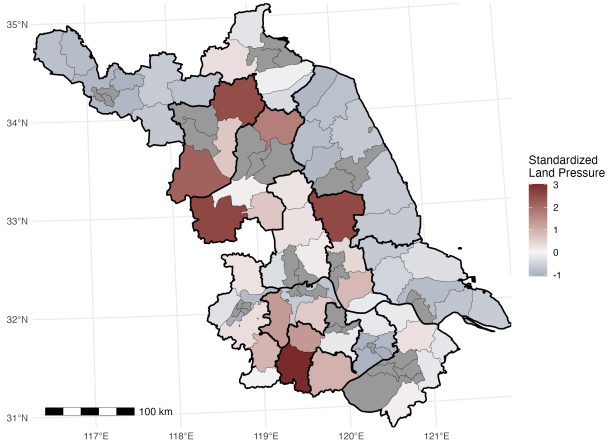
Back to main figure

Data

Variable	Proxy	Data
Remote-sensing and raster data		
Cropland	Land cover	CLCD, 30 m annual land cover
Cultivation intensity	Vegetation cover	FVC, 30 m / 15-day Fractional Vegetation Cover
Quality	Soil type	1995 soil data and 2023 soil data
Administrative and economic data		
Officials		Public biographies of prefecture- and province-level officials
Boundaries / controls		Mainland China county map; County Statistical Yearbooks
Planning obligations		Prefecture Land Use Master Plans (2006–2020)
Land transaction		Jiangsu Land Market transaction records
Growth pressure	GDP growth targets, Q1 GDP growth	Government Work Reports

Back to outcomes

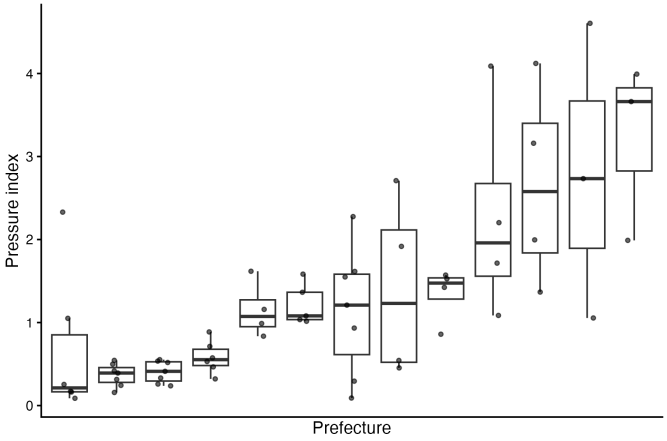
Standardized Land Protection Pressure in Jiangsu



Note: Positive values = above-average exposure; gray = core urban districts.

Back to measurement

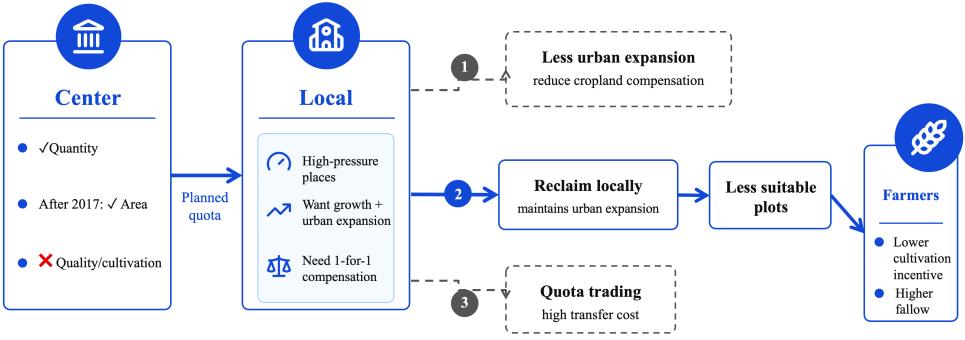
Distribution of Cropland Protection Pressure within and across Prefectures



Notes. The unstandardized pressure index is shown for non-core counties. Boxes indicate the interquartile range within each prefecture, thick lines indicate medians, and dots indicate individual counties.

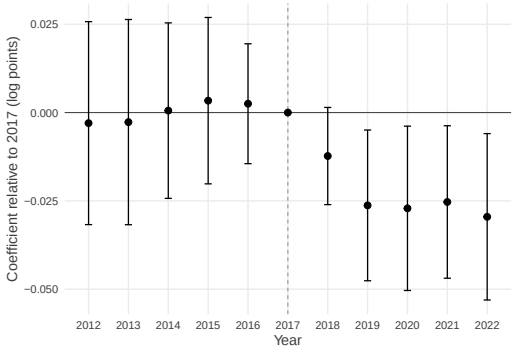
Back to measurement

Conceptual Framework: From Imperfect Monitoring to Misallocation



Back to identification

Cultivation Intensity: Crop Output per Cropland Hectare

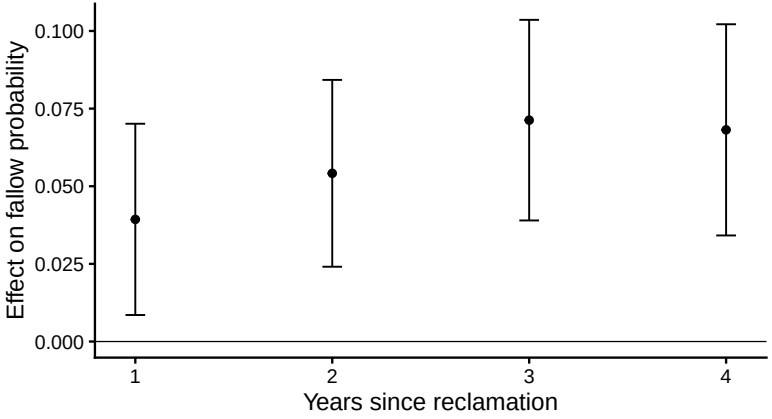


- Before 2017, crop output per hectare shows **no differential trend** ($p = 0.741$).
- From 2019–2022, a 1 s.d. increase in pressure implies **2.5–3.0% lower output per hectare** ($p < 0.05$ each year).

County-year event study; grain + oilseed + cotton output per satellite-measured cropland hectare; county and year fixed effects; standard errors clustered by county.

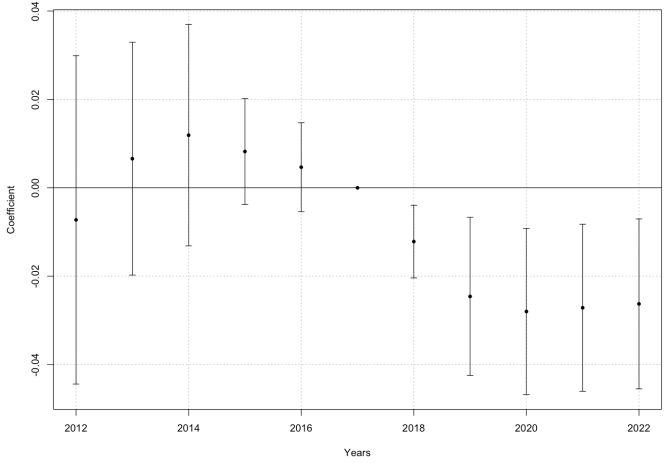
Back to outcomes

Cultivation Intensity: Fallow Event Study



Back to cultivation intensity

Event Study of FVC



Back to robustness check

Alternative Definition of Land Pressure

Table: Robustness: Alternative Definitions of Land-Pressure Exposure

	(1) Cropland Share	(2) Soil suitability	(3) Fallow
Panel A: All Land Types as Available Land			
Pressure × Post	0.006^{***}	-0.071^{***}	0.078^{***}
	(0.002)	(0.024)	(0.026)
Grid & Year FE	✓		
County & Cohort FE		✓	✓
Years-since-reclamation FE			✓
Cluster level	County	County	County
Number of counties	64	52	54
Num. Obs.	3,743,982	939	3,243
R^2	0.981	0.452	0.230
Panel B: Alternative Definition of Core-Urban Districts			
Pressure × Post	0.003^{**}	-0.051^{*}	0.084^{**}
	(0.002)	(0.028)	(0.032)
Grid & Year FE	✓		
County & Cohort FE		✓	✓
Years-since-reclamation FE			✓
Cluster level	County	County	County
Number of counties	64	52	54
Num. Obs.	3,743,982	939	3,243
R^2	0.981	0.451	0.234

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

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Sensitivity Analysis

- Sensitivity to Reclaimed-Land Classification
  - Baseline: Reclaimed$_t = 1$ if $\{CropShare_{t-1} < 0.60\} \wedge \{CropShare_t, CropShare_{t+1} > 0.70\}$, where 0.70 is the pre-treatment mean.
  - Robustness: $0.60 \rightarrow 0.55$–0.65 for the previous year's threshold.
- Sensitivity to Fallow Definition
  - Baseline: Fallow $= 1$ if FVC < 0.20, where 0.20 corresponds to the bottom 25%.
  - Robustness: $0.20 \rightarrow 0.13$–0.20.

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Sensitivity to Alternative Thresholds in Reclaimed-Land Definition

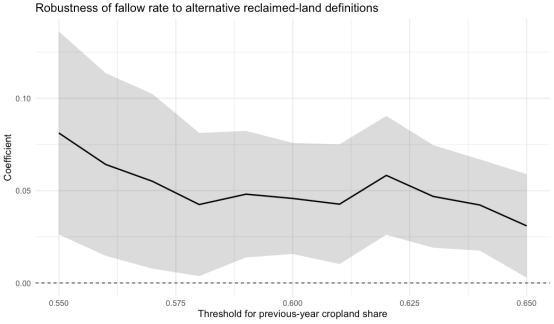


Figure: Fallow

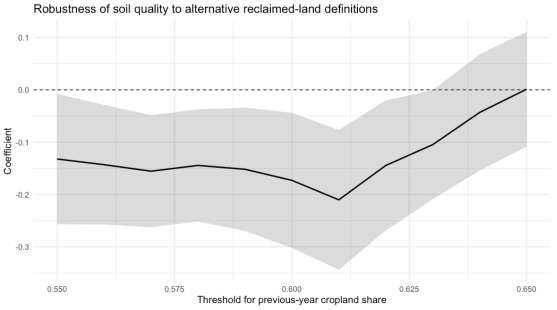
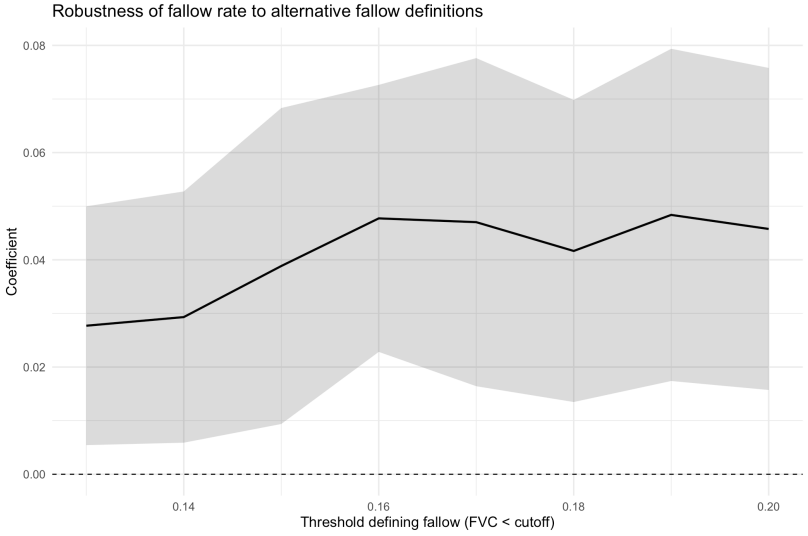


Figure: Soil Quality

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Sensitivity to Alternative Fallow Rate Definition



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Placebo Test

	(1) Triangle	(2) Epanech.	(3) Uniform
Panel A: Δ Cropland share			
RD in 2013−2012	−0.0016	−0.0016	−0.0016
(Jiangsu − Anhui)	(0.0027)	(0.0026)	(0.0026)
RD in 2014−2012	−0.0007	−0.0005	−0.0004
(Jiangsu − Anhui)	(0.0034)	(0.0033)	(0.0033)
Panel B: Δ Fallow rate			
RD in 2013−2012	0.0112	0.0097	0.0073
(Jiangsu − Anhui)	(0.0093)	(0.0100)	(0.0100)
RD in 2014−2012	0.0018	−0.0005	−0.0086
(Jiangsu − Anhui)	(0.0157)	(0.0158)	(0.0161)

⇒ No pre-policy discontinuity at the border.

Notes. Placebo RD-DID assigning the policy to pre-policy years (2013, 2014); the outcome is the Δ cropland share (or fallow rate) relative to 2012. No pre-policy estimate is statistically significant.

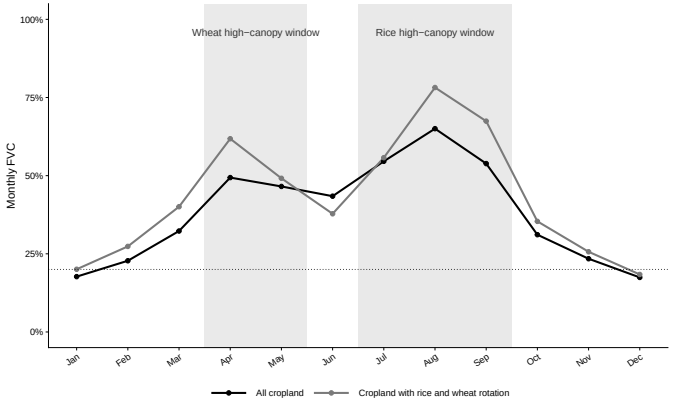
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Additional Results: Spatial Allocation and Environmental Outcomes

	Spatial allocation			Conversion to cropland		
	Dist. to existing cropland (m) (1)	Zero-cropland grids (2)	Isolated grids (3)	Water (4)	Forest (5)	Wetland (6)
Pressure × Post-2017	4.222**	0.470**	0.263**	0.082***	-0.010	0.038***
	(1.782)	(0.184)	(0.123)	(0.018)	(0.008)	(0.009)
	[0.035]	[0.032]	[0.045]	[< 0.001]	[0.238]	[< 0.001]
Magnitudes:						
1-SD Pressure	4.662**	0.519**	0.291**	0.090***	-0.011	0.042***
Outcome mean	11.011	1.189	0.311	0.214	0.046	0.087
County FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Observations	640	640	640	576	576	576
R^2	0.336	0.324	0.200	0.690	0.478	0.587
County clusters	64	64	64	64	64	64
Pre-policy joint p-value	0.317	0.600	0.288	0.101	0.366	0.335

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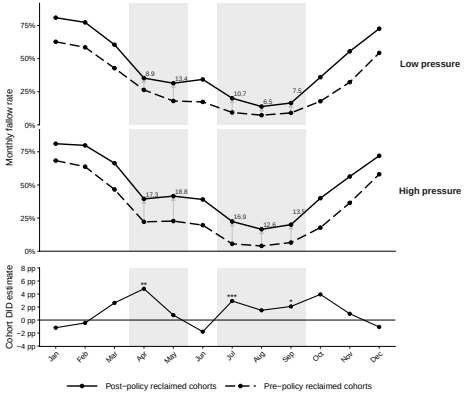
Normal Seasonality: Cultivation Calendar



Finding. April–May and July–September are the high-canopy growing windows in both all cropland and rice–wheat rotation fields.

Cohort-DiD estimates
Back to alternative explanations

Normal Seasonality: Month-Specific Fallow Estimates

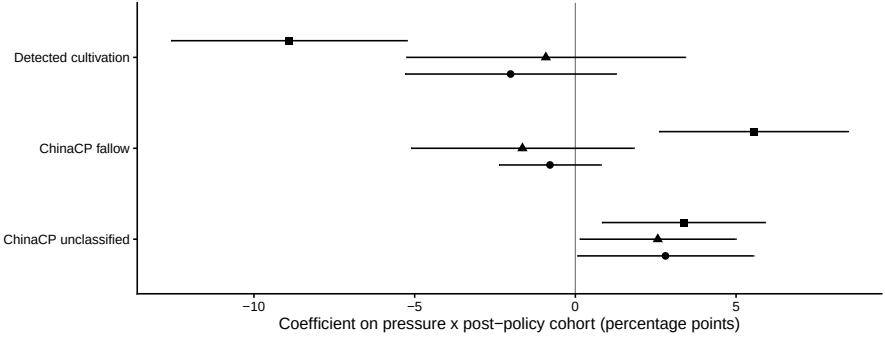


Finding. In high-pressure counties, the cohort-DiD estimates are positive throughout the April–May and July–September growing windows.

Cultivation calendar

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Land Preparation: Appendix Evidence

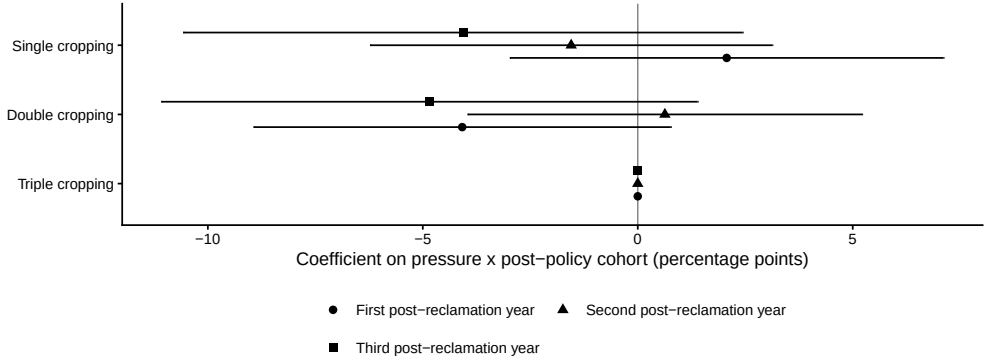


Years after reclamation: ● Year 1 ▲ Year 2 ■ Year 3

Finding. If preparation drove the result, the fallow gap would peak immediately after reclamation. Instead, there is no clear difference in years 1–2; in year 3, detected cultivation falls by **8.9 p.p.** and ChinaCP fallow rises by **5.5 p.p.**

◂ Back to alternative explanations

Crop Switching: Appendix Evidence



Finding. Among detected cultivation, pressure does not clearly change single-, double-, or triple-cropping shares. Nor is any crop-specific estimate significant after false-discovery-rate adjustment.

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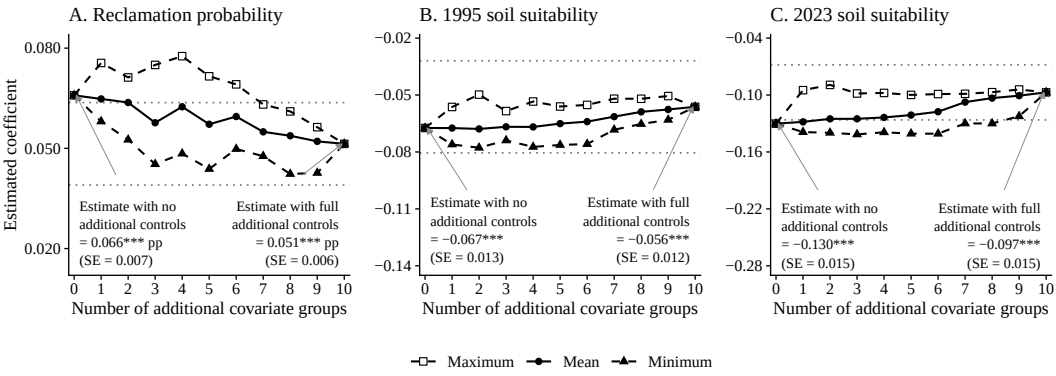
Mechanism: Monitoring-Station Distance and Reclamation

	Extensive margin				Intensive margin: Suitable share			
	1(Reclaimed in year)				1995 soil data		2023 soil data	
	Level gradient		Grid FE					
	Baseline (1)	+ Urban (2)	Baseline (3)	+ Urban (4)	Baseline (5)	+ Urban (6)	Baseline (7)	+ Urban (8)
$\log(d_i) \times \text{Post}$	0.080^{***}	0.066^{***}	0.080^{***}	0.066^{***}	0.010	0.011	−0.004	−0.006
	(0.007)	(0.007)	(0.007)	(0.007)	(0.014)	(0.014)	(0.016)	(0.016)
$\log(d_i)$	0.011^{**}	0.014^{***}			-0.064^{***}	-0.067^{***}	-0.135^{***}	-0.130^{***}
	(0.004)	(0.004)			(0.013)	(0.013)	(0.015)	(0.015)
Grid FE			✓	✓				
Year FE	✓	✓	✓	✓				
County FE	✓	✓			✓	✓	✓	✓
Reclamation-cohort FE					✓	✓	✓	✓
Baseline land-use controls	✓	✓			✓	✓	✓	✓
Urban-center distance control		✓		✓		✓		✓
Outcome mean (share)	0.002	0.002	0.002	0.002	0.742	0.742	0.623	0.623
Observations	15,512,250	15,512,250	15,512,250	15,512,250	30,891	30,891	30,892	30,892
R^2	0.007	0.007	0.101	0.101	0.345	0.346	0.407	0.408
Cluster level	Parent 500 m grid							

Notes. d_i: distance from 100 m grid i to the nearest pre-2017 provincial cropland-quality monitoring station. Cols. (1)–(4): grid-years 2013–2022, Post = 2018–2022, coefficients in p.p. Cols. (5)–(8): grids first reclaimed in 2013–2022. Soil outcomes: area-weighted shares of source pixels/components with score ≥ 3, inherited from parent 500 m grids. SEs clustered by parent grid; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Coef. stability | Back to mechanism | Back to figure

Mechanism: Coefficient Stability of Monitoring-Distance Estimates



- Add subsets of **ten covariate groups**: county-boundary and government-seat distance; soil suitability, cropland, impervious, forest, and water shares within 5 km of the station; elevation; ruggedness; nighttime lights.
- (a) $\log(d_i)\times$Post for annual reclamation (p.p.); (b)–(c) level coefficient on $\log(d_i)$ for 1995 soil data / 2023 soil data suitable shares. Lines: mean, max, min across subsets; dotted: 95% CI of the ten-group specification.
- **Signs are retained across all combinations; fully adjusted estimates remain significant.**

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Mechanism: Reclamation Costs and Land Suitability

	Low-cost, unsuitable		High-cost, suitable	
	1995 soil data	2023 soil data	1995 soil data	2023 soil data
	(1)	(2)	(3)	(4)
Land type × Post × Pressure	2.514***	2.153***	-0.700***	-0.783***
	(0.832)	(0.774)	(0.175)	(0.208)
Land type × Post	1.333	1.800*	-0.906	-0.962*
	(1.082)	(0.970)	(0.552)	(0.563)
Post × Pressure	0.590*	0.594*	1.037***	1.084***
	(0.330)	(0.323)	(0.348)	(0.355)
Additional effect per 1-SD pressure	2.776***	2.377***	-0.773***	-0.864***
	(0.918)	(0.854)	(0.193)	(0.229)
Pixel FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Observations	60,743,493	61,414,714	60,743,493	61,414,714
R^2	0.332	0.333	0.332	0.333
Counties	64	64	64	64
Cluster level	County	County	County	County

Notes. Land type is low-cost, unsuitable in (1)–(2) and high-cost, suitable in (3)–(4). Prior cultivation proxies for lower reclamation cost; no cultivation history proxies for higher cost. Historical cultivation: at least two consecutive cropland years in 2001–2011; all pixels are noncropland in 2012. Suitability uses the score cutoff of three. Outcome: $100 \times \mathbf{1}$(first persistent reclamation); coefficients in p.p. Post: 2018 onward. Pixel-years, 2013–2021. County-clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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Mechanism: Frontier Component vs. Selection Wedge

Selection accounts for 73–78% of the suitability decline.

Reference predictors	Total loss	Frontier component	Selection wedge	Frontier share (%)	Selection share (%)
Remaining suitable share	10.156	2.413	7.743	23.8	76.2
Remaining suitable share + depletion	10.156	2.204	7.952	21.7	78.3
Remaining suitable share + depletion + reclamation intensity	10.156	2.718	7.438	26.8	73.2

- **Reference**: 2002–2017 relation between each cohort's quality and the county's remaining reservoir, applied to 2018–2021 with actual quantities held fixed.
- **Frontier component**: historical mean − reference (costly compliance).
- **Selection wedge**: reference − observed quality (proxy for distortion).

Notes. 92 county–cohort observations in 37 counties, reclaimed-area weighted; suitable-land share (p.p.) under the 1995 soil map. Back-of-the-envelope decomposition.

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Mechanism: Economic Growth Pressure

- **Growth burden**: target growth minus predicted annual growth based on quarter 1; high burden when positive and in the top tercile in Jiangsu that year.

	Cropland share (1)	New construction land area (ha) (2)	Cropland-origin construction area (ha) (3)
Pressure × Post × growth burden	0.006^{*}	267.707^{**}	261.115^{**}
	(0.003)	(95.222)	(86.557)
	[0.358]	[0.340]	[0.301]
Party-secretary controls	✓	✓	✓
Grid FE	✓		
Prefecture FE	✓	✓	✓
Year FE	✓	✓	✓
Observations	3,050,309	116	116
R^2	0.985	0.604	0.637
Prefecture clusters	13	13	13

Notes. Pressure: prefecture cropland protection pressure; Post: after 2017. Col. (1): non-core rural cropland; cols. (2)–(3): urban-core conveyances signed after March 31. Cropland origin uses prior-year cropland at the transaction centroid. All lower-order interactions included. Parentheses: prefecture-clustered SEs; brackets: wild-cluster bootstrap p-values (9,999 draws). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

◂ Officials' incentives ◂ Back to mechanism

Agricultural Losses and Local Incentives

- A production function maps suitability and fallowing effects into agricultural value-added losses.
- A one-standard-deviation increase in pressure implies an **11.1% annual loss**.

	Agricultural baseline	Primary-industry baseline
Annual loss (RMB/ha/year)	3,456	5,530
Capitalized perpetual loss (RMB/ha)	59,583	95,346
Share of conveyance revenue	0.14%	0.23%

- Nearby new construction-land conveyances yield **RMB 42.1 million/ha** in gross revenue.

Notes. All values in 2017 RMB. Capitalization holds annual losses constant. Revenue is total deflated payments divided by area for new construction-land conveyances within 5 km of land reclaimed in the same year.

◂ Officials' incentives ◂ Back to mechanism

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